Sulforaphane potentiates oxaliplatin-induced cell growth inhibition in colorectal cancer cells via induction of different modes of cell death.

The objective of this study was to investigate, whether the plant-derived isothiocyanate Sulforaphane (SFN) enhances the antitumor activities of the chemotherapeutic agent oxaliplatin (Ox) in a cell culture model of colorectal cancer. Caco-2 cells were cultured under standard conditions and treated with increasing concentrations of SFN [1-20 μM] and/or Ox [100 nM-10 μM]. For co-incubation, cells were pre-treated with SFN for 24 h. Cell growth was determined by BrdU incorporation. Drug interactions were assessed using the combination-index method (CI) (Cl < 1 indicates synergism). Apoptotic events were characterized by different ELISA techniques. Protein levels were examined by Western blot analysis. Annexin V- and propidium iodide (PI) staining followed by FACS analysis was used to differentiate between apoptotic and necrotic events. SFN and Ox alone inhibited cell growth of Caco-2 cells in a dose-dependent manner, an effect, which could be synergistically enhanced, when cells were incubated with the combination of both agents. Co-treated cells further displayed distinctive morphological changes that occurred during the apoptotic process, such as cell surface exposure of phosphatidylserine, membrane blebbing as well as the occurence of cytoplasmic histone-associated DNA fragments. Further observations thereby pointed toward simultaneous activation of both extrinsic and intrinsic apoptotic pathways. With increasing concentrations and treatment duration, a shift from apoptotic to necrotic cell death could be observed. In conclusion, the data suggest that the isothiocyanate SFN sensitizes colon cancer cells to Ox-induced cell growth inhibition via induction of different modes of cell death.